

SOFTWARE REQUIREMENTS SPECIFICATION

CSI-100 Software System (IVD)

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Purpose

This specification defines the functional and non-functional requirements for the CSI-100 software system, its software items, interfaces, safety classification and performance criteria, per IEC 62304 clause 5.2. It is the binding input for software architecture, detailed design, and unit and integration testing.

Document Control

Rev	Date	Software version	Description of change	Author
A	Jan 2025	0.5	Initial release for architecture phase.	A. Lodhi
B	Jun 2025	1.0	Added fourth beam spot, 561/445 nm laser support and configuration management.	A. Lodhi
C	Nov 2025	2.0	Added plate deposition, index sorting and aerosol option control.	A. Lodhi
D	Feb 2026	2.5	Added IVD features (roles, audit trail, electronic signatures, assay lock) and cybersecurity requirements.	A. Lodhi
E	Aug 2026	3.0	Restructured into four parts: Research Application, Firmware, Admin Application and Data Management Application. Added firmware board management, fail-safe, internal log and workstation communication requirements; added hardware configuration control and measurement and calibration requirements; aligned with SRS-CSI-001 Rev D; traceability generated from the requirement set.	A. Lodhi

Approvals

Function	Role	Signature / Date
Software Engineering	Software Lead (applications)	
Firmware Engineering	Firmware and FPGA Lead	
Systems Engineering	Author, System Requirements Owner	
Software Quality Assurance	Software Quality	
Quality Engineering	Risk Management	
Service & Support	Admin Application user representative	
Regulatory Affairs	Regulatory Review	

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for the CSI-100 software system. It meets IEC 62304 clause 5.2 (software requirements analysis) and is the binding input for software architecture design (SAD-SW-CSI-011), detailed design, and unit, integration and software system testing.

Every requirement traces up to a system requirement in SRS-CSI-001 or to a risk control in HA-CSI-001, and down to the software item that implements it and the tests that verify it. The traceability tables in Section 11 are generated from the requirement tables in this document, so both directions are complete.

1.2 Scope and document structure

The software system consists of three workstation applications, embedded firmware, and two shared workstation services. This document is organized in four parts, one for each major software product, followed by requirements common to all of them:

Part	Software product	Users	Main purpose
A	Research Application (desktop)	Operators, supervisors	Run the instrument: fluidics, acquisition, analysis, sort setup, sorting, daily QC and IVD assays
B	Firmware	None directly (runs on the instrument)	Real-time sorting management, FPGA board control, fail-safe, internal log, communication with the workstation
C	Admin Application	Administrators, service engineers, manufacturing	Hardware configuration control; measurement and calibration of lasers, optics, fluidics and sort system; user, assay and bead lot administration
D	Data Management Application	Supervisors, administrators	Records, review and release, audit trail, electronic signatures, export, backup, archive and retention
Common	Shared services and requirements	All	Security, cybersecurity, performance, reliability, platform, data and SOUP

This document covers software release 3.0 for all laser and detector configurations in CFG-CSI-001. It does not cover third-party analysis software, laboratory information systems, or the internal firmware of purchased components (lasers, PMT modules, water bath), which are treated as SOUP where the CSI-100 software depends on them.

1.3 Requirement conventions

Requirement identifiers show the part they belong to: RA-*nnn* (Research Application), FW-*nnn* (Firmware), AD-*nnn* (Admin Application), DM-*nnn* (Data Management Application) and CM-*nnn* (common). Identifiers are never reused.

Shall marks a mandatory, verified requirement. Each requirement table gives:

Attribute	Meaning
Parent	System requirement (SYS- <i>nnn</i> , SRS-CSI-001) or hazard risk control (HA- <i>nn</i> , HA-CSI-001)
Item	Software item(s) that implement the requirement (Section 3.2)
Class	IEC 62304 safety class of the implementing item(s) (Section 2)
Test level	Lowest level at which the requirement is verified: U = unit, I = integration, S = software system test

1.4 Lifecycle context

The software is developed under the Software Development Plan (SDP-CSI-010) following the IEC 62304 lifecycle: requirements analysis (this document), architectural design, detailed design, unit implementation and verification, integration and integration testing, software system testing, and release. Software risk management follows ISO 14971 and IEC 62304 clause 7. Changes after release follow the Software Configuration Management Plan (SCMP-CSI-010) and include re-evaluation of safety classification and traceability.

2. Software Safety Classification

2.1 Software system classification

The CSI-100 software system is **Class B** under IEC 62304 clause 4.3. A software failure could contribute to a hazardous situation: an incorrect result from a misclassified population or wrong configuration, an incorrect sorted fraction, or failure to stop the stream and turn off high voltage after a clog. After external risk control measures, the resulting harm is judged non-serious:

- Laser exposure is prevented by a hardware shutter interlock on the flow cell access door, independent of software.
- High voltage is removed by the hardwired emergency stop circuit and the sort block door interlock, independent of software.
- Overpressure is prevented by mechanical relief valves on the pressurized tanks.
- Aerosol containment relies on sealed doors, evacuation hardware and ULPA filtration; software adds a secondary response.
- Incorrect clinical results are mitigated by assay controls, daily bead-based QC, result review by qualified staff and interpretation with other clinical findings.

No software item is Class C. If a change removes or weakens any of these external controls, the classification is re-evaluated before release.

2.2 Software item classification

Part	Item	Name	Class
A. Research Application	SI-01	Operator UI and Workflow	B
A. Research Application	SI-02	Acquisition and Analysis	B
A. Research Application	SI-03	Sort Control	B
A. Research Application	SI-04	Fluidics and Instrument Control	B
A. Research Application	SI-05	QC Run and Assay Execution	B
B. Firmware	SI-10	Instrument Controller Firmware	B
B. Firmware	SI-11	Sort Logic FPGA	B
B. Firmware	SI-12	Board Management FPGA	B
B. Firmware	SI-13	Stream Imaging Firmware	B
B. Firmware	SI-14	Fluidics Module Firmware	A
B. Firmware	SI-15	Safety Supervisor Firmware	B
C. Admin Application	SI-20	Hardware Configuration Control	B
C. Admin Application	SI-21	Measurement and Calibration Tools	B
C. Admin Application	SI-22	User, Assay and Bead Lot Administration	B
D. Data Management Application	SI-30	Data Management Application	B
D. Data Management Application	SI-31	Database and Records Service	B
D. Data Management Application	SI-32	Audit Trail and E-Signature Service	B
Shared	SI-40	Security Service	B
Shared	SI-41	Instrument Communication Library	B

SI-14 (Class A) runs on a separate processor in the fluidics supply module; it communicates with SI-10 only through the serial interface in ICD-CSI-010, so its failure cannot corrupt Class B items. SI-15 runs on an independent microcontroller with its own watchdog so that fail-safe actions do not depend on SI-10.

3. Software System Overview

3.1 Context

The operator runs the instrument from the Research Application on the workstation. Real-time functions (pulse processing, sort decisions, drop charging, stream imaging, board control and fail-safe) run in firmware on the instrument. The Admin Application is used by administrators, service engineers and manufacturing to control the hardware configuration and to measure and calibrate lasers, optics, fluidics and the sort system. The Data Management Application holds all records and provides review, release, audit, export, backup and archive. All three applications use the shared Security Service for login and roles, and the shared Instrument Communication Library to talk to the firmware.

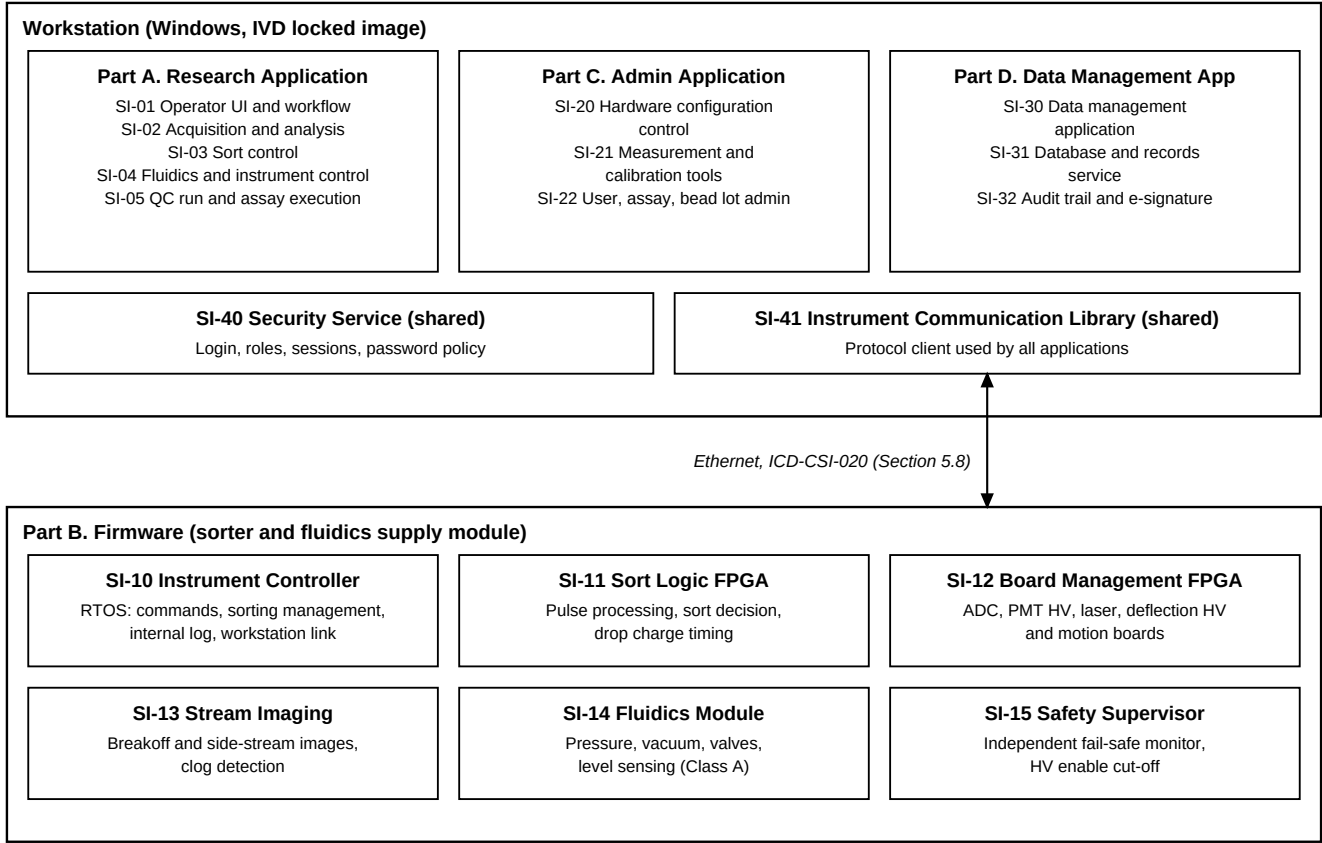


Figure 1. CSI-100 software system: four parts, software items and main interfaces.

3.2 Software items

Item	Name	Platform	Main responsibilities
SI-01	Operator UI and Workflow	Workstation	Workspace, plots, dialogs, guided procedures, alarms and messages
SI-02	Acquisition and Analysis	Workstation	Parameters, display scaling, compensation, gating, statistics
SI-03	Sort Control	Workstation	Sort setups, stream and breakoff control, drop delay, layouts, precision modes, sort reports
SI-04	Fluidics and Instrument Control	Workstation	Startup, shutdown, cleaning, sample loading, agitation, temperature, options
SI-05	QC Run and Assay Execution	Workstation	Daily performance check run, QC lockout, IVD assay template execution
SI-10	Instrument Controller Firmware	Sorter, RTOS	Command execution, sorting management, motion, lasers, internal log, workstation link
SI-11	Sort Logic FPGA	Sorter, FPGA	Digitization, pulse height/area/width, laser delay, sort decision, charge timing
SI-12	Board Management FPGA	Sorter boards, FPGA/CPLD	Board identification, ADC, PMT high voltage, laser, deflection HV and motion board control and monitoring
SI-13	Stream Imaging Firmware	Sorter, embedded	Breakoff and side-stream image capture and analysis, clog detection
SI-14	Fluidics Module Firmware	Fluidics module MCU	Compressor, pressure regulation, vacuum, valves, level sensing
SI-15	Safety Supervisor Firmware	Sorter, safety MCU	Independent monitoring of heartbeats and fault signals; drives the safe state
SI-20	Hardware Configuration Control	Workstation	Controlled record of installed lasers, optics, detectors, fluidics and options
SI-21	Measurement and Calibration Tools	Workstation	Laser, optics, detector, fluidics, sort system and stage measurement and calibration
SI-22	User, Assay and Bead Lot Administration	Workstation	Users and roles, assay template approval, bead lots, default setups
SI-30	Data Management Application	Workstation	Record browser, review and release, export, backup, archive
SI-31	Database and Records Service	Workstation	Transactional storage of all records
SI-32	Audit Trail and E-Signature Service	Workstation	Audit trail capture and electronic signatures for all applications
SI-40	Security Service	Workstation	Authentication, roles, sessions, password policy
SI-41	Instrument Communication Library	Workstation	Workstation side of the instrument protocol

3.3 External interfaces

Interface	Between	Definition
IF-01 Workstation to sorter	SI-41 and SI-10	Dedicated Ethernet link; command, telemetry and event data (ICD-CSI-020, Section 5.8)
IF-02 Sorter to fluidics module	SI-10 and SI-14	Serial link in the umbilical (ICD-CSI-010)
IF-03 Controller to boards	SI-10 and SI-12	Internal board bus; register map in SDS-FW-CSI-013
IF-04 Safety signals	SI-15, SI-10, hardware	Heartbeat lines, HV enable, charge enable, interlock and e-stop inputs
IF-05 Operator	SI-01, SI-30, Admin	Keyboard, mouse, one or two monitors, printer
IF-06 Data export	SI-30	FCS 3.1, PDF and CSV files; optional LIS result export
IF-07 Calibration references	SI-21	Manual entry or USB import of readings from calibrated external instruments (power meter, pressure gauge, balance)

3.4 Roles and application access

Function	Operator	Supervisor	Administrator	Service
Research Application: run, acquire, sort, daily QC	Yes	Yes	Yes	Yes
Research Application: create research templates	Yes	Yes	Yes	No
Research Application: QC lockout override (research use only)	No	Yes	Yes	No
Admin Application: view configuration and calibration	No	Yes	Yes	Yes
Admin Application: change configuration, calibrate	No	No	Yes	Yes
Admin Application: users, bead lots, assay approval	No	Approve assays	Yes	No
Data Management: view own records	Yes	Yes	Yes	No
Data Management: review, release, audit trail	No	Yes	Yes	No
Data Management: backup, archive, retention, deletion	No	No	Yes	No
Firmware diagnostics and internal log export	No	No	Yes	Yes

PART A. RESEARCH APPLICATION

The Research Application is the desktop application operators use to run the CSI-100. It covers fluidics procedures, acquisition and analysis, sort setup and stream control, sorting, the daily performance check and the execution of research templates and locked IVD assay templates.

4. Research Application Requirements

4.1 Workflow and user interface

ID	Requirement	Parent	Item	Class	Test level
RA-101	The application shall provide a workspace with experiment browser, plots, acquisition controls, sort controls, fluidics status and alarm panel.	SYS-901	SI-01	B	S
RA-102	The application shall guide the operator through startup, daily QC, sorting and shutdown with step-by-step instructions and completion status.	SYS-901	SI-01	B	S
RA-103	The application shall show fluid levels, system pressure, laser status, HV status, active configuration, QC status and connection status at all times.	SYS-493, SYS-522	SI-01	B	S
RA-104	The application shall ask for confirmation before any action that discards unsaved data or stops an active sort.	SYS-904	SI-01	B	S
RA-105	The application shall show progress and remaining time for each automated procedure.	SYS-901	SI-01	B	S

4.2 Instrument and fluidics control

ID	Requirement	Parent	Item	Class	Test level
RA-201	The application shall provide an automated fluidics startup that pressurizes the system, primes sheath, sample and waste lines and removes air, and reports completion or the step that failed.	SYS-601	SI-04	B	S
RA-202	The application shall provide an automated fluidics shutdown that fills the flow path with 70% ethanol using the closed-loop nozzle and depressurizes the system.	SYS-602	SI-04	B	S
RA-203	The application shall provide cleaning modes: flow cell clean, sample line backflush, prime after tank refill, and a bleach, deionized water and ethanol clean cycle.	SYS-603	SI-04	B	S
RA-204	The application shall not start fluidics startup unless a nozzle is detected as installed and the flow cell door is closed.	HA-09	SI-04, SI-10	B	I
RA-205	On Load, the application shall raise the loading port, pressurize the sample chamber and start flow at the selected rate; on Unload it shall stop flow, lower the port and flush the sample line inside and outside.	SYS-472, SYS-494	SI-04, SI-10	B	I
RA-206	The application shall offer flow rate settings 1 to 11 (about 10 to 80 $\mu\text{L}/\text{min}$), sample agitation speeds and Off, and sample chamber temperature setpoints and Off.	SYS-453, SYS-495, SYS-4101	SI-04	B	S
RA-207	The application shall warn when sheath falls below 20% or waste rises above 80%, and shall block a sort when capacity is not enough for the planned sort.	SYS-493	SI-04	B	S
RA-208	The application shall detect the aerosol evacuation module, temperature control option and plate deposition unit, enable their functions only when detected, and warn if sorting starts while the evacuation module is connected but not running.	SYS-803, SYS-551	SI-04	B	S

4.3 Acquisition and analysis

ID	Requirement	Parent	Item	Class	Test level
RA-301	The application shall let the user set detector voltages, FSC gain, threshold parameter and level, and window extension, and select height, area and width for each parameter.	SYS-443	SI-02	B	S
RA-302	The application shall display data on a linear scale from 2.6 to 262,144 or on a four- or five-decade log scale.	SYS-445	SI-02	B	U
RA-303	The application shall acquire and display at least 70,000 events per second with 8 parameters and 12 compensation pairs, and more than 100,000 events per second with fewer parameters, without losing events.	SYS-451, SYS-452	SI-02, SI-41	B	S
RA-304	The application shall report events lost to electronic abort or overflow and warn the user when any occur.	SYS-451	SI-02	B	I
RA-305	The application shall calculate a compensation matrix from single-stained controls and apply it to acquisition and analysis.	SYS-632	SI-02	B	U
RA-306	The application shall let users define polygon, rectangle, interval and quadrant gates, hierarchies and statistics, and apply them in real time.	SYS-631	SI-02	B	S
RA-307	The application shall store acquired data with all instrument settings, configuration version and QC status needed to reproduce the analysis.	SYS-635, SYS-705	SI-02, SI-31	B	I

4.4 Sort setup and stream control

ID	Requirement	Parent	Item	Class	Test level
RA-401	The application shall load the controlled default sort setup for the selected nozzle (70, 85, 100 or 130 μ m), as released in the Admin Application (AD-701), and provide one editable custom setup per nozzle.	SYS-611	SI-03	B	S
RA-402	Changing the sort setup shall require confirmation of the installed nozzle size.	HA-08	SI-03	B	S
RA-403	The application shall restore the last setup at startup with stream, breakoff control, plate voltage, test sort and optical filter always Off.	SYS-611, HA-02	SI-03	B	I
RA-404	The application shall display the breakoff and side-stream images at least 10 times per second.	SYS-612	SI-03, SI-13	B	I
RA-405	The application shall let the user set drop drive frequency (1.0 to 102.0 kHz), amplitude (1.0 to 80.0 V), strobe phase (0 to 360°), attenuation, drop delay (10 to 140 drops), stream charges, center voltage, plate voltage and 2nd to 4th drop corrections (–100% to 100%).	SYS-463, SYS-611	SI-03	B	S
RA-406	The application shall let the user turn plate voltage on and off, run a test sort, move the drop-delay optical filter and open or close the aspirator drawer.	SYS-614	SI-03	B	S
RA-407	The application shall determine drop delay automatically with drop-delay beads by sweeping drop delay and selecting the value with the highest deflected fraction; the result shall be within ± 0.25 drop of the manual method.	SYS-465	SI-03, SI-13	B	S

4.5 Automatic breakoff control and clog handling

ID	Requirement	Parent	Item	Class	Test level
RA-501	When breakoff control is On, the application shall adjust drop drive amplitude to bring Drop 1 to its target in coarse then fine steps until the Gap target is met, and shall disable manual amplitude and frequency entry.	SYS-613	SI-03, SI-13	B	I
RA-502	If Gap differs from its target by more than 1 pixel, the application shall adjust amplitude to restore it.	SYS-613	SI-03	B	I
RA-503	The application shall record each sort pause and resume commanded by firmware (FW-205), with cause and duration, in the sort record.	SYS-635	SI-03	B	I
RA-504	After a clog, the application shall show the clog recovery procedure, including running aerosol evacuation in boost mode before opening the sort block door when the option is installed.	SYS-554, HA-04	SI-01	B	S
RA-505	The application shall not allow the stream or plate voltage to restart after a clog until the operator acknowledges the clog alarm.	HA-04	SI-03, SI-10	B	I

4.6 Sort layouts, collection devices and precision modes

ID	Requirement	Parent	Item	Class	Test level
RA-601	The application shall provide sort layouts for two-way 12 × 75 mm and 15 mL tubes; four-way 12 × 75 mm, 1.5 mL and 1 mL tubes; 6-, 24-, 48-, 96- and 384-well plates; and slides; plus custom devices released in the Admin Application.	SYS-481, SYS-482, SYS-485	SI-03	B	S
RA-602	The application shall let the user assign gated populations to each sort location, set target counts and save several layouts per tube.	SYS-631	SI-03	B	S
RA-603	The application shall not allow a four-way layout with a two-way device, or a plate layout without the plate deposition unit.	HA-08	SI-03	B	U
RA-604	The application shall provide the read-only precision modes in Table 1 and any additional modes released in the Admin Application, and send the selected yield, purity and phase masks to firmware.	SYS-464	SI-03	B	S
RA-605	Yield and phase masks shall be mutually exclusive: setting one above 0 shall set the other to 0.	SYS-464	SI-03	B	U

Precision mode	Yield mask	Purity mask	Phase mask	Single cell	Use
Purity	32	32	0	No	Default two-way sorting
4-Way Purity	0	32	0	No	Four-way sorting
Yield	32	0	0	No	Enrichment
Single Cell	0	32	16	Yes	Plate and slide sorting, counting
Initial	32	0	0	No	Drop delay setup
Fine Tune	0	0	0	No	Drop delay fine tuning

Table 1. Default precision modes.

4.7 Sort execution, plate deposition and sort report

ID	Requirement	Parent	Item	Class	Test level
RA-701	The application shall not start a sort unless the stream is on, breakoff control is on, plate voltage is on, the performance check has passed (RA-801) and the collection device matches the layout.	SYS-622, HA-08	SI-03, SI-05	B	S
RA-702	The application shall display, for each sort location, sorted count, sort rate, conflict count and sort efficiency, updated at least once per second.	SYS-633	SI-01, SI-03	B	S
RA-703	The user shall be able to pause, resume and stop a sort; stopping shall close the sort record without losing counts.	SYS-634	SI-03	B	I
RA-704	The application shall stop the sort automatically when all target counts are reached.	SYS-633	SI-03	B	S
RA-705	In index sorting, the application shall store the parameters of each deposited event and its well with the sort record.	SYS-484	SI-03, SI-31	B	I
RA-706	At the end of each sort the application shall create a sort report with operator, date and time, instrument serial number and configuration version, nozzle and setup, precision mode, drop delay, layout, populations, target and achieved counts, conflicts, pauses and causes, and QC status.	SYS-635	SI-03, SI-31	B	S

4.8 Daily performance check and QC lockout

ID	Requirement	Parent	Item	Class	Test level
RA-801	The application shall run the bead-based performance check against the baseline released in the Admin Application (AD-601), measuring each enabled detector (robust CV, intensity, voltage) and reporting Pass, Pass with warning or Fail per parameter.	SYS-621	SI-05	B	S
RA-802	The performance check shall set inter-laser delay and area scaling automatically and apply them to acquisition.	SYS-444, SYS-621	SI-05, SI-11	B	I
RA-803	The application shall not allow acquisition or sorting with an IVD assay template unless a performance check for the active configuration has passed in the previous 24 hours.	SYS-622, HA-08	SI-05	B	S
RA-804	A supervisor shall be able to record a documented QC lockout override for research use only; results acquired under override shall be marked not for diagnostic use.	SYS-622	SI-05, SI-40	B	S
RA-805	The application shall warn the user before acquisition if the active lasers do not match the selected configuration.	SYS-623, HA-08	SI-05, SI-10	B	I
RA-806	The application shall display trends of detector voltage, robust CV and laser delay with baseline and limits.	SYS-624	SI-05	B	S

4.9 IVD assay execution

ID	Requirement	Parent	Item	Class	Test level
RA-901	When running an approved IVD assay template, the application shall lock instrument settings, parameters, compensation, gates, statistics and report layout; operators shall not be able to change them.	SYS-704, HA-08	SI-05, SI-02	B	S
RA-902	Results from an IVD assay template shall show the template name and version and whether QC passed.	SYS-704, SYS-635	SI-05	B	S
RA-903	The application shall mark any result acquired while a QC, configuration or fluidics warning was active.	HA-08	SI-02, SI-05	B	S
RA-904	The application shall send completed IVD results to the Data Management Application for review and release; results shall not be reportable until released (DM-301).	SYS-703	SI-05, SI-31	B	I

4.10 Alarms and operator messages

ID	Requirement	Parent	Item	Class	Test level
RA-1001	The application shall classify alarms as high, medium and low, and show each with its cause and the action to take.	SYS-901	SI-01	B	S
RA-1002	High alarms shall include clog, emergency stop, loss of instrument communication, sheath empty, waste full, fluidics fault, HV fault and any fail-safe entry reported by firmware.	HA-04, HA-06, HA-15	SI-01, SI-04	B	S
RA-1003	The application shall show the HV-on indicator whenever firmware reports plate voltage enabled.	SYS-522, HA-02	SI-01	B	I

PART B. FIRMWARE

Firmware runs on the instrument and performs every time-critical and safety-related function: sorting management, FPGA code that manages the instrument boards, fail-safe behavior, the internal log and communication with the workstation. Firmware must keep the instrument safe even if the workstation stops responding.

5. Firmware Requirements

5.1 Firmware structure and boot

ID	Requirement	Parent	Item	Class	Test level
FW-101	On power-up, SI-10 shall run a power-on self test of memory, clocks, board bus and safety supervisor, and shall report the result to the workstation before accepting operating commands.	SYS-1007	SI-10	B	I
FW-102	Each FPGA shall load its configuration only after verifying its CRC; a failed check shall hold that board in reset and report a board fault.	HA-08	SI-11, SI-12	B	U
FW-103	Firmware shall start in the safe state (Section 5.6) and leave it only on command from an authenticated workstation.	HA-02	SI-10, SI-15	B	I
FW-104	Firmware and FPGA images shall be accepted for update only if their digital signature is valid and they are listed as compatible in the release manifest; a failed update shall leave the previous version running.	SYS-707, SYS-708	SI-10	B	I

5.2 Sorting management

ID	Requirement	Parent	Item	Class	Test level
FW-201	SI-11 shall compute height, area and width for every enabled parameter of every event, and apply per-laser delay and area scaling so that signals from each beam spot align to the same event.	SYS-443, SYS-444	SI-11	B	U
FW-202	SI-11 shall digitize signals on an 18-bit linear scale and process at least 70,000 events per second with 8 parameters without loss, and more than 100,000 events per second with fewer parameters.	SYS-445, SYS-451, SYS-452	SI-11	B	U
FW-203	SI-11 shall resolve the position of each event within its drop to 1/32 of a drop and apply the yield, purity and phase masks (0 to 32) received from the Research Application as defined in SDS-FW-CSI-013.	SYS-464	SI-11	B	U
FW-204	SI-11 shall complete the sort decision for each event within 20 μ s of the end of its event window, and schedule the drop charge at the correct drop and phase for the active drop delay, with deterministic timing independent of event rate.	SYS-461	SI-11	B	U
FW-205	During sorting, if Gap differs from its target by more than 2 pixels or Drop 1 by more than 10 pixels, SI-11 shall stop charging within 100 ms while sample continues to flow, and resume automatically when both are in range.	SYS-542, HA-07	SI-11, SI-13	B	I
FW-206	Single-cell modes shall sort at most one target event per drop and count every sorted event exactly once.	SYS-483	SI-11	B	U
FW-207	Pause shall stop charging within 100 ms while acquisition continues; resume shall restart charging at the next drop boundary.	SYS-634	SI-11	B	I
FW-208	SI-10 shall move the plate stage to each well in layout order and enable sorting only when the stage is in position.	SYS-482, SYS-483	SI-10	B	I
FW-209	SI-13 shall capture strobed breakoff images at least 10 times per second and compute Drop 1 (100 to 600 pixels) and Gap for every image.	SYS-612, SYS-613	SI-13	B	U
FW-210	SI-13 shall detect a clog (loss or deflection of the stream, or Gap above the setup upper limit) within 500 ms.	SYS-541, HA-04	SI-13	B	U

5.3 FPGA board management

ID	Requirement	Parent	Item	Class	Test level
FW-301	Each instrument board (ADC, PMT high voltage, laser control, deflection HV, charge, motion and camera boards) shall carry an identity record with part number, serial number, hardware revision and calibration data.	SYS-1005	SI-12	B	I
FW-302	At startup, SI-10 shall read every board identity, compare the inventory with the released hardware configuration (AD-101), and report missing, extra or incompatible boards; a mismatch shall block operation with IVD templates.	SYS-623, HA-08	SI-10, SI-12	B	I
FW-303	SI-12 shall configure the ADC boards (sample rate, gain, offset) from the board calibration data at startup.	SYS-443	SI-12	B	U
FW-304	SI-12 shall set each PMT high voltage from 0 to 1,000 V as commanded, read it back, and report a fault if the readback differs from the setpoint by more than 1% for more than 1 second.	SYS-441, HA-08	SI-12	B	U
FW-305	SI-12 shall switch each laser on and off, set laser power where the laser supports it, and read back laser power, current, temperature and interlock status at least once per second.	SYS-433, SYS-435	SI-12	B	U
FW-306	SI-12 shall enforce that lasers sharing a beam spot (488/445 and 405/375) are never on at the same time.	SYS-435	SI-12	B	U
FW-307	SI-12 shall set deflection plate voltage and stream charge levels as commanded, read back plate voltage, and report a fault if the readback differs by more than 2%.	SYS-523, HA-02	SI-12	B	U
FW-308	SI-12 shall monitor supply rails (within $\pm 5\%$), board temperatures and fan speeds, and report any value outside its limits.	HA-12	SI-12	B	U
FW-309	SI-12 shall drive the loading port, aspirator drawer, drop-delay filter and plate stage motors with position feedback and stall detection.	SYS-562	SI-12	B	U
FW-310	SI-12 shall expose board status and counters through the register map defined in SDS-FW-CSI-013, readable by SI-10 at least 10 times per second.	SYS-1007	SI-12	B	I

5.4 Fluidics module firmware

ID	Requirement	Parent	Item	Class	Test level
FW-401	SI-14 shall hold sheath pressure within $\pm 1\%$ of the setpoint across 5 to 75 PSI and report pressure to SI-10 at least 10 times per second.	SYS-491	SI-14	A	U
FW-402	SI-14 shall report sheath, waste, ethanol and cleaning fluid levels and supply air pressure; supply pressure outside 80 to 100 PSI for more than 5 seconds with the stream on shall be reported as a fluidics fault.	SYS-493	SI-14, SI-10	B	I
FW-403	On main power off, SI-14 shall vent system pressure within 10 seconds.	SYS-524	SI-14	A	I

5.5 Sample handling and interlocks

ID	Requirement	Parent	Item	Class	Test level
FW-501	The loading port shall move only while the emergency stop is not active, and SI-10 shall light the emergency stop button while the port is moving.	SYS-533	SI-10	B	I
FW-502	SI-10 shall monitor door interlock, laser shutter and emergency stop status and reject stream, plate voltage and loading port commands that conflict with them.	SYS-512, SYS-521	SI-10	B	I

5.6 Fail-safe

The safe state is: deflection plate voltage off, drop charging off, sample tube unloaded, loading port and plate stage stopped, aspirator drawer closed, and the stream either stopped (clog, emergency stop) or left running (other faults, to avoid drying the nozzle). Lasers remain as set; laser exposure is controlled by the hardware shutter interlock.

ID	Requirement	Parent	Item	Class	Test level
FW-601	On emergency stop, SI-10 shall unload the sample tube, stop the plate stage, turn off stream and plate voltage and close the aspirator drawer within 2 seconds, and report the event. (The hardware circuit removes HV independently.)	SYS-532, HA-02, HA-06	SI-10	B	I
FW-602	On a detected clog, SI-10 shall, without waiting for the workstation, stop the stream, turn off plate voltage, close the aspirator drawer and unload the sample tube within 2 seconds of detection.	SYS-543, HA-04	SI-10	B	I
FW-603	If communication with the workstation is lost for more than 2 seconds during sorting, SI-10 shall stop charging, turn off plate voltage and keep the stream running.	SYS-706, HA-02	SI-10	B	I
FW-604	SI-15 shall monitor heartbeats from SI-10 and SI-11; if either is missing for more than 250 ms, SI-15 shall remove the HV enable and charge enable signals in hardware.	HA-02	SI-15	B	U
FW-605	SI-10 and SI-15 shall each run a hardware watchdog; a watchdog timeout shall put the instrument in the safe state.	HA-02	SI-10, SI-15	B	U
FW-606	A board fault reported under FW-304, FW-307 or FW-308, an FPGA configuration failure or an over-temperature shall put the instrument in the safe state.	HA-02, HA-08, HA-12	SI-10, SI-12	B	I
FW-607	The safe state shall be latched: firmware shall leave it only after the cause has cleared and an authenticated user has acknowledged the fault.	HA-02, HA-04	SI-10	B	I
FW-608	SI-10 shall report HV status at least 5 times per second and report every entry into the safe state with its cause.	SYS-522	SI-10	B	I

5.7 Internal log

ID	Requirement	Parent	Item	Class	Test level
FW-701	SI-10 shall keep an internal log of events, warnings, faults, safe-state entries, commands that change instrument state, configuration changes, firmware versions and power cycles.	SYS-1007	SI-10	B	I
FW-702	Each log entry shall contain a UTC time stamp with millisecond resolution synchronized to the workstation, severity, source item and a coded message.	SYS-1007	SI-10	B	U
FW-703	The log shall be kept in non-volatile memory for at least 100,000 entries or 90 days, whichever is larger, as a ring buffer; entries written up to 1 second before a power loss shall be preserved.	SYS-1007	SI-10	B	I
FW-704	SI-10 shall keep usage counters for laser on-hours per laser, PMT on-hours, stream hours, sort hours, compressor hours and power cycles.	SYS-1001, SYS-1005	SI-10	B	U
FW-705	The log shall contain no patient or specimen data, and shall be readable and exportable only through the Admin Application by administrator or service roles (AD-901).	SYS-1007, SYS-709	SI-10	B	I
FW-706	Log entries shall not be editable or deletable by any user; the log export shall include a checksum.	SYS-702	SI-10	B	U

5.8 Workstation communication

ID	Requirement	Parent	Item	Class	Test level
FW-801	The instrument shall communicate with the workstation over a dedicated Ethernet link using the protocol in ICD-CSI-020, with separate channels for commands, telemetry and event data.	SYS-802	SI-10, SI-41	B	I
FW-802	The instrument shall accept connections only from the paired workstation, authenticated at connection time.	SYS-707	SI-10, SI-41	B	I
FW-803	Every message shall carry a sequence number and a CRC-32; messages with a bad CRC, a sequence error or out-of-range parameters shall be rejected without changing instrument state and logged.	SYS-707, HA-08	SI-10	B	U
FW-804	The instrument shall acknowledge every command within 100 ms with accepted, rejected (with reason) or completed status.	SYS-706	SI-10	B	U
FW-805	Instrument and workstation shall exchange heartbeats every 250 ms; loss for more than 2 seconds shall trigger FW-603 and a high alarm in the Research Application.	SYS-706	SI-10, SI-41	B	I
FW-806	The event data channel shall carry at least 110,000 events per second at 8 parameters without loss, and the instrument shall buffer at least 2 seconds of event data if the link is briefly slow.	SYS-451, SYS-452	SI-10, SI-11, SI-41	B	I
FW-807	Workstation and instrument shall check protocol version compatibility at connection and refuse to operate on a mismatch.	HA-08	SI-10, SI-41	B	I
FW-808	The instrument clock shall be synchronized to the workstation at connection and at least hourly.	SYS-702	SI-10, SI-41	B	I

PART C. ADMIN APPLICATION

The Admin Application is used by administrators, service engineers and manufacturing. It controls the hardware configuration of each instrument and provides the measurement and calibration functions for lasers, optics and detectors, fluidics, the sort system and the plate stage. It also administers users, assay template approval, bead lots and default sort setups. It is not used for routine operation.

6. Admin Application Requirements

6.1 Hardware configuration control

ID	Requirement	Parent	Item	Class	Test level
AD-101	The application shall maintain a controlled hardware configuration record for each instrument: installed lasers (wavelength, serial number, rated power), detector arrays and PMTs (position, serial number), mirrors and filters per position, FSC ND filter, nozzles, fluidics module serial number and installed options.	SYS-623, SYS-1008	SI-20	B	S
AD-102	The application shall compare the configuration record with the board inventory reported by firmware (FW-302) and show every difference.	SYS-623, HA-08	SI-20, SI-10	B	I
AD-103	Each change to the configuration record shall require the administrator or service role, a reason, and an electronic signature, and shall create a new configuration version; previous versions shall be retained.	SYS-623, SYS-703	SI-20, SI-32	B	S
AD-104	After a configuration change affecting lasers, optics or detectors, the application shall require a new performance check baseline (AD-601) before IVD templates can run.	SYS-622, HA-08	SI-20, SI-05	B	S
AD-105	The application shall enforce the configuration rules in CFG-CSI-001, including: unique fluorochrome names per laser; the 405 nm array bandpass set to 480/20 when the 445 nm laser is active; and the mirror and filter changes required by the 3-PMT 633 nm and 5-PMT 561 nm options.	SYS-623	SI-20	B	U
AD-106	The application shall generate the optical configuration (parameter names, detector positions, mirrors, filters) used by the Research Application from the configuration record.	SYS-623	SI-20	B	I
AD-107	The application shall record field upgrades of lasers and detector arrays as configuration changes, with the upgrade kit number.	SYS-1008	SI-20	B	S

6.2 Laser measurement and calibration

ID	Requirement	Parent	Item	Class	Test level
AD-201	The application shall display, for each laser, measured power, current, temperature, on-hours and interlock status from firmware (FW-305, FW-704).	SYS-433, SYS-1007	SI-21	B	S
AD-202	The application shall provide a laser power calibration that records power measured at the laser head and at the flow cell with a calibrated external power meter, and stores the calibration factor for the laser's internal monitor.	SYS-433	SI-21	B	S
AD-203	The application shall flag any laser whose power at the head is below the minimum in SRS-CSI-001 (SYS-433) and block IVD templates that use it.	SYS-433, HA-08	SI-21, SI-05	B	S
AD-204	The application shall provide a warm-up verification that records laser power stability over 30 minutes after power-on.	SYS-437	SI-21	B	S
AD-205	The application shall calibrate inter-laser delay and area scaling for each laser and nozzle with alignment beads, and store the results as the default values used by the performance check.	SYS-444	SI-21	B	S

6.3 Optics and detector measurement and calibration

ID	Requirement	Parent	Item	Class	Test level
AD-301	The application shall provide an alignment verification tool that runs alignment beads and reports robust CV, intensity and signal position for FSC, SSC and each primary detector, against the limits in FTS-CSI-001.	SYS-436	SI-21	B	S
AD-302	The application shall calibrate each PMT's voltage-to-gain curve with multi-intensity beads and store the curve with the PMT serial number.	SYS-441	SI-21	B	S
AD-303	The application shall verify detector linearity with multi-peak beads and report the deviation from linear response for each detector.	SYS-411	SI-21	B	S
AD-304	The application shall provide a resolution check with chicken erythrocyte nuclei that reports the G0/G1 CV and doublet/singlet ratio against SYS-401, SYS-402 and SYS-411.	SYS-401, SYS-402, SYS-411	SI-21	B	S
AD-305	The application shall record the installed FSC ND filter and apply the matching FSC scaling.	SYS-423	SI-21, SI-20	B	S

6.4 Fluidics measurement and calibration

ID	Requirement	Parent	Item	Class	Test level
AD-401	The application shall calibrate each pressure sensor at two points against a calibrated reference gauge and store the offset and gain in the fluidics module.	SYS-491	SI-21, SI-14	B	I
AD-402	The application shall calibrate sample flow rate by recording the sample volume consumed (weighed on a calibrated balance) over a timed run at each flow rate setting 1 to 11, and store the resulting table.	SYS-453	SI-21	B	S
AD-403	The application shall calibrate each level sensor at empty and full.	SYS-493	SI-21, SI-14	B	I
AD-404	The application shall run a pressure decay leak test of the sheath and sample paths and report pass or fail against the limit in FTS-CSI-001.	SYS-491	SI-21	B	S
AD-405	The application shall calibrate the sample chamber temperature sensor against a reference thermometer and store the offset.	SYS-4101	SI-21	B	S

6.5 Sort system and stage measurement and calibration

ID	Requirement	Parent	Item	Class	Test level
AD-501	The application shall calibrate breakoff and side-stream camera scale and focus with the camera calibration target and store pixels per millimetre.	SYS-612	SI-21, SI-13	B	I
AD-502	The application shall verify drop drive frequency and plate voltage against readback and against an external meter, and store correction factors.	SYS-463, SYS-523	SI-21, SI-12	B	I
AD-503	The application shall measure stream position in the waste aspirator for each nozzle and guide the service engineer through sort block pivot adjustment.	SYS-468	SI-21	B	S
AD-504	The application shall calibrate the plate stage home position and the first and last well of each collection device, and verify single-bead deposition accuracy against SYS-483.	SYS-483	SI-21, SI-12	B	S
AD-505	The application shall verify collection temperature control against a reference thermometer when the option is installed.	SYS-4102	SI-21	B	S

6.6 Performance check baselines and bead lots

ID	Requirement	Parent	Item	Class	Test level
AD-601	The application shall create a performance check baseline for each optical configuration at installation, after service and after configuration changes, and store it with date, bead lot, configuration version and user.	SYS-621	SI-22, SI-21	B	S
AD-602	The application shall import bead lot files with target values and limits, and reject files that are expired or fail their integrity check.	SYS-806, HA-08	SI-22	B	U
AD-603	The application shall manage drop-delay bead and alignment bead lots with expiry dates.	SYS-806	SI-22	B	S

6.7 Default sort setups, custom devices and precision modes

ID	Requirement	Parent	Item	Class	Test level
AD-701	The application shall hold the default sort setups for each nozzle as controlled records containing the values in SSD-CSI-700 §6; changes shall require the administrator role and electronic signature, and create a new version.	SYS-611	SI-22, SI-32	B	S
AD-702	The application shall let an administrator define custom collection devices by rows, columns, spacing and home position.	SYS-485	SI-22	B	S
AD-703	The application shall let an administrator create, name and delete additional precision modes; default modes shall be read-only.	SYS-464	SI-22	B	S

6.8 Users and assay templates

ID	Requirement	Parent	Item	Class	Test level
AD-801	The application shall let an administrator create, disable and assign roles to user accounts; accounts shall never be deleted.	SYS-701	SI-22, SI-40	B	S
AD-802	The application shall let a supervisor approve an IVD assay template by electronic signature; any change shall create a new version that requires approval before use.	SYS-704, SYS-703	SI-22, SI-32	B	S

6.9 Calibration records and firmware diagnostics

ID	Requirement	Parent	Item	Class	Test level
AD-901	The application shall display firmware diagnostics (pressures, laser status, voltages, temperatures, board status, internal log) and export the internal log, for administrator and service roles only, without access to patient data.	SYS-1007	SI-21	B	S
AD-902	Each calibration shall be recorded with as-found and as-left values, pass or fail, reference instrument ID and its calibration due date, user, date and time.	SYS-1006	SI-21, SI-31	B	S
AD-903	The application shall warn when a calibration is due within 30 days and block IVD templates when a calibration that affects results is overdue.	SYS-1004, HA-08	SI-21, SI-05	B	S
AD-904	The application shall produce a calibration certificate report for each instrument.	SYS-1006	SI-21	B	S
AD-905	Calibration tools used in manufacturing shall produce records in the format required by FTS-CSI-001 for the Device History Record.	SYS-1301	SI-21	B	S

PART D. DATA MANAGEMENT APPLICATION

The Data Management Application holds all records created by the Research and Admin Applications. It provides record browsing, review and release of results, the audit trail, electronic signatures, export, backup, archive and retention. The Research and Admin Applications write through the same Database and Records Service.

7. Data Management Application Requirements

7.1 Records and browsing

ID	Requirement	Parent	Item	Class	Test level
DM-101	The application shall let users browse and search experiments, tubes, sort records, QC results and calibration records by date, user, instrument, template and specimen ID.	SYS-635	SI-30	B	S
DM-102	The application shall display stored data, plots and statistics without an instrument connected.	SYS-635	SI-30	B	S
DM-103	SI-31 shall use transactions so each record is written completely or not at all, and shall allow the Research Application to write while the Data Management Application reads.	SYS-706	SI-31	B	U
DM-104	On workstation power loss or crash, data recorded up to the last completed second, sort records and the audit trail shall remain intact after restart.	SYS-706	SI-31	B	S
DM-105	The application shall warn the administrator when database storage exceeds 80% of its allocated space.	SYS-706	SI-31, SI-30	B	S

7.2 Audit trail

ID	Requirement	Parent	Item	Class	Test level
DM-201	SI-32 shall record every creation, modification and deletion of experiments, data, templates, configurations, calibrations, users and settings from any application, with user ID, date, time, old value, new value and reason where required.	SYS-702	SI-32	B	I
DM-202	Audit trail entries shall be generated by the software within 1 second of the action, time-stamped from a synchronized clock, and not editable or deletable by any user.	SYS-702	SI-32	B	U
DM-203	Supervisors shall be able to view, filter and export the audit trail.	SYS-702	SI-30	B	S
DM-204	Security events (logins, failed logins, account changes, update attempts) shall be recorded in the audit trail.	SYS-702, SYS-708	SI-32, SI-40	B	S

7.3 Review, release and electronic signatures

ID	Requirement	Parent	Item	Class	Test level
DM-301	IVD results shall have status Draft, Reviewed or Released; only released results can be printed as reports or exported to a LIS.	SYS-703	SI-30	B	S
DM-302	Electronic signatures shall require re-entry of the password and record the signer's name, date, time and meaning of the signature.	SYS-703	SI-32	B	S
DM-303	Signatures shall be linked to their records so that they cannot be copied or transferred to another record.	SYS-703	SI-32	B	U
DM-304	Changing a released record shall create a new version with reason and signature; the released version shall be retained.	SYS-703	SI-30, SI-32	B	S

7.4 Export and reports

ID	Requirement	Parent	Item	Class	Test level
DM-401	The application shall export data in FCS 3.1 format with all keywords needed to reproduce the acquisition, reports in PDF and statistics in CSV.	SYS-705	SI-30	B	U
DM-402	Exported files shall include a checksum so that later changes can be detected.	SYS-709	SI-30	B	U
DM-403	The application shall offer an export option that removes patient identifiers.	SYS-709	SI-30	B	S
DM-404	The application shall export released IVD results to a laboratory information system in a configurable CSV or HL7 v2 format.	SYS-705	SI-30	B	S

7.5 Backup, archive and retention

ID	Requirement	Parent	Item	Class	Test level
DM-501	The application shall provide scheduled and manual backup of the database and data files, and restore with verification of completeness and checksums.	SYS-702	SI-30	B	S
DM-502	The application shall archive records to external storage with integrity checks, and retrieve archived records with their audit trail and signatures.	SYS-702	SI-30	B	S
DM-503	Records shall be retained for the periods in Section 9; deletion shall be possible only for an administrator, with reason, as a logged soft delete.	SYS-702	SI-30, SI-32	B	S
DM-504	User data at rest shall be encrypted and accessible only through the applications.	SYS-709	SI-31	B	I

COMMON REQUIREMENTS

8. Security, Cybersecurity and Non-Functional Requirements

8.1 Security service

ID	Requirement	Parent	Item	Class	Test level
CM-101	All applications shall require login with a unique user ID and password through SI-40 before any function is available.	SYS-701	SI-40	B	S
CM-102	SI-40 shall provide operator, supervisor, administrator and service roles with the permissions in Section 3.4.	SYS-701	SI-40	B	S
CM-103	Passwords shall have configurable length, complexity and expiry; accounts shall lock after a configurable number of failed attempts; sessions shall lock after a configurable idle time.	SYS-701, SYS-709	SI-40	B	U

8.2 Cybersecurity

ID	Requirement	Parent	Item	Class	Test level
CM-201	The workstation image shall enable only the services and ports needed for operation; the instrument link shall be isolated from other networks.	SYS-707, SYS-802	SI-41	B	S
CM-202	Application updates shall be accepted only if their digital signature is valid.	SYS-707, SYS-708	SI-30	B	I
CM-203	A software bill of materials covering applications and firmware shall be generated for every release.	SYS-708	SI-10, SI-30	B	S

8.3 Performance, reliability and platform

ID	Requirement	Parent	Item	Class	Test level
CM-301	Applications shall update plots at least twice per second at the maximum event rate and respond to user input within 0.5 seconds.	SYS-901	SI-01	B	S
CM-302	Each application shall start and, where needed, connect to the instrument within 60 seconds.	SYS-604	SI-01, SI-41	B	S
CM-303	The software system shall run 24 hours of continuous acquisition at the maximum event rate without memory growth that affects performance.	SYS-1002	SI-02, SI-10	B	S
CM-304	Applications shall run on the validated workstation configuration in the release notes and check operating system and database versions at startup.	SYS-707	SI-30	B	S
CM-305	User interface text shall be stored in resource files to support translation, and the user interfaces shall follow the style guide in SDS-SW-CSI-012.	SYS-1105, SYS-902	SI-01	B	I

9. Data Requirements

Data item	Created by	Retention and protection
Experiment and tube records	Research Application	Until deleted by an administrator; deletion audited
Sort records, sort reports, index sort data	Research Application	Retained with the experiment
IVD results and signatures	Research and Data Management	At least 10 years or as required by the laboratory
Performance check results and baselines	Research and Admin	At least 7 years
Hardware configuration versions	Admin Application	Life of the instrument
Calibration records and certificates	Admin Application	Life of the instrument
Default setups, devices, precision modes, assay templates	Admin Application	All versions retained
Audit trail	All applications	At least 10 years; not editable
User accounts	Admin Application	Disabled, never deleted
Firmware internal log	Firmware	Ring buffer on the instrument (FW-703); exports retained with service records

10. SOUP Requirements

Software of unknown provenance is listed with its purpose and requirements (IEC 62304 clauses 5.3.3 and 7.1.2). Known anomalies are reviewed for each version before release.

SOUP	Used by	Purpose	Requirement
Windows 10 IoT Enterprise LTSC	Workstation applications	Operating system	Supported for product life; updates validated before release
.NET runtime (LTS)	SI-01 to SI-41 (workstation)	Application runtime	LTS release; no runtime installed from the internet
Embedded SQL database engine	SI-31	Data storage	Transactional writes; encrypted database files
Plotting library	SI-01, SI-02, SI-30	Data display	Render 1,000,000 events per plot at ≥ 2 updates per second
Cryptographic library	SI-40, SI-10, SI-30	Hashing, signatures, TLS	FIPS 140-validated algorithms
Real-time operating system	SI-10	Task scheduling	Deterministic scheduling; certified kernel
FPGA vendor IP cores	SI-11, SI-12	ADC interface, DSP blocks, bus	Timing closure at design clock
Image processing library	SI-13	Breakoff image analysis	Process ≥ 10 frames per second

11. Verification and Traceability

11.1 Verification approach

Requirements are verified at the lowest practical level: unit tests (U) for requirements within one software unit; integration tests (I) for requirements that depend on interaction between items or with hardware; software system tests (S) on a production-equivalent instrument. Verification is planned in SVP-CSI-010 and reported in SVR-SW-CSI-010.

Level	Scope	Test document	Environment
Unit	Units within each software item	UTS-CSI-010 series (one per item)	Automated framework in CI; FPGA simulation for SI-11 and SI-12
Integration	Item to item, software to hardware	ITS-CSI-010	Hardware-in-the-loop bench and integration instrument
Software system	Complete software system	STS-CSI-010 (Parts A to D and common)	Production-equivalent instrument, all configurations
Regression	Changed and affected areas	Regression suite in STS-CSI-010	Every release candidate

11.2 Upward traceability: parent requirements to software requirements

Generated from the Parent column of every requirement table. Every software requirement appears under at least one parent.

Parent	Software requirements
SYS-401	AD-304
SYS-402	AD-304
SYS-411	AD-303, AD-304
SYS-423	AD-305
SYS-433	FW-305, AD-201, AD-202, AD-203
SYS-435	FW-305, FW-306
SYS-436	AD-301
SYS-437	AD-204
SYS-441	FW-304, AD-302
SYS-443	RA-301, FW-201, FW-303
SYS-444	RA-802, FW-201, AD-205
SYS-445	RA-302, FW-202
SYS-451	RA-303, RA-304, FW-202, FW-806
SYS-452	RA-303, FW-202, FW-806
SYS-453	RA-206, AD-402
SYS-461	FW-204
SYS-463	RA-405, AD-502
SYS-464	RA-604, RA-605, FW-203, AD-703
SYS-465	RA-407
SYS-468	AD-503
SYS-472	RA-205

Parent	Software requirements
SYS-481	RA-601
SYS-482	RA-601, FW-208
SYS-483	FW-206, FW-208, AD-504
SYS-484	RA-705
SYS-485	RA-601, AD-702
SYS-491	FW-401, AD-401, AD-404
SYS-493	RA-103, RA-207, FW-402, AD-403
SYS-494	RA-205
SYS-495	RA-206
SYS-512	FW-502
SYS-521	FW-502
SYS-522	RA-103, RA-1003, FW-608
SYS-523	FW-307, AD-502
SYS-524	FW-403
SYS-532	FW-601
SYS-533	FW-501
SYS-541	FW-210
SYS-542	FW-205
SYS-543	FW-602
SYS-551	RA-208
SYS-554	RA-504
SYS-562	FW-309
SYS-601	RA-201
SYS-602	RA-202
SYS-603	RA-203
SYS-604	CM-302
SYS-611	RA-401, RA-403, RA-405, AD-701
SYS-612	RA-404, FW-209, AD-501
SYS-613	RA-501, RA-502, FW-209
SYS-614	RA-406
SYS-621	RA-801, RA-802, AD-601
SYS-622	RA-701, RA-803, RA-804, AD-104
SYS-623	RA-805, FW-302, AD-101, AD-102, AD-103, AD-105, AD-106
SYS-624	RA-806
SYS-631	RA-306, RA-602
SYS-632	RA-305
SYS-633	RA-702, RA-704
SYS-634	RA-703, FW-207

Parent	Software requirements
SYS-635	RA-307, RA-503, RA-706, RA-902, DM-101, DM-102
SYS-701	AD-801, CM-101, CM-102, CM-103
SYS-702	FW-706, FW-808, DM-201, DM-202, DM-203, DM-204, DM-501, DM-502, DM-503
SYS-703	RA-904, AD-103, AD-802, DM-301, DM-302, DM-303, DM-304
SYS-704	RA-901, RA-902, AD-802
SYS-705	RA-307, DM-401, DM-404
SYS-706	FW-603, FW-804, FW-805, DM-103, DM-104, DM-105
SYS-707	FW-104, FW-802, FW-803, CM-201, CM-202, CM-304
SYS-708	FW-104, DM-204, CM-202, CM-203
SYS-709	FW-705, DM-402, DM-403, DM-504, CM-103
SYS-802	FW-801, CM-201
SYS-803	RA-208
SYS-806	AD-602, AD-603
SYS-901	RA-101, RA-102, RA-105, RA-1001, CM-301
SYS-902	CM-305
SYS-904	RA-104
SYS-1001	FW-704
SYS-1002	CM-303
SYS-1004	AD-903
SYS-1005	FW-301, FW-704
SYS-1006	AD-902, AD-904
SYS-1007	FW-101, FW-310, FW-701, FW-702, FW-703, FW-705, AD-201, AD-901
SYS-1008	AD-101, AD-107
SYS-1105	CM-305
SYS-1301	AD-905
SYS-4101	RA-206, AD-405
SYS-4102	AD-505
HA-02	RA-403, RA-1003, FW-103, FW-307, FW-601, FW-603, FW-604, FW-605, FW-606, FW-607
HA-04	RA-504, RA-505, RA-1002, FW-210, FW-602, FW-607
HA-06	RA-1002, FW-601
HA-07	FW-205
HA-08	RA-402, RA-603, RA-701, RA-803, RA-805, RA-901, RA-903, FW-102, FW-302, FW-304, FW-606, FW-803, FW-807, AD-102, AD-104, AD-203, AD-602, AD-903
HA-09	RA-204
HA-12	FW-308, FW-606
HA-15	RA-1002

11.3 Downward traceability: software items to requirements and tests

Generated from the Item column of every requirement table.

Item	Requirements implemented	Unit tests	Integration / system tests
SI-01	RA-101, RA-102, RA-103, RA-104, RA-105, RA-504, RA-702, RA-1001, RA-1002, RA-1003, CM-301, CM-302, CM-305	UTS-CSI-010-01	ITS §1; STS §A
SI-02	RA-301, RA-302, RA-303, RA-304, RA-305, RA-306, RA-307, RA-901, RA-903, CM-303	UTS-CSI-010-02	ITS §2; STS §A
SI-03	RA-401, RA-402, RA-403, RA-404, RA-405, RA-406, RA-407, RA-501, RA-502, RA-503, RA-505, RA-601, RA-602, RA-603, RA-604, RA-605, RA-701, RA-702, RA-703, RA-704, RA-705, RA-706	UTS-CSI-010-03	ITS §3; STS §A
SI-04	RA-201, RA-202, RA-203, RA-204, RA-205, RA-206, RA-207, RA-208, RA-1002	UTS-CSI-010-04	ITS §4; STS §A
SI-05	RA-701, RA-801, RA-802, RA-803, RA-804, RA-805, RA-806, RA-901, RA-902, RA-903, RA-904, AD-104, AD-203, AD-903	UTS-CSI-010-05	ITS §5; STS §A
SI-10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-605, FW-606, FW-607, FW-608, FW-701, FW-702, FW-703, FW-704, FW-705, FW-706, FW-801, FW-802, FW-803, FW-804, FW-805, FW-806, FW-807, FW-808, AD-102, CM-203, CM-303	UTS-CSI-010-10	ITS §6; STS §B
SI-11	RA-802, FW-102, FW-201, FW-202, FW-203, FW-204, FW-205, FW-206, FW-207, FW-806	UTS-CSI-010-11	ITS §7; STS §B
SI-12	FW-102, FW-301, FW-302, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309, FW-310, FW-606, AD-502, AD-504	UTS-CSI-010-12	ITS §8; STS §B
SI-13	RA-404, RA-407, RA-501, FW-205, FW-209, FW-210, AD-501	UTS-CSI-010-13	ITS §9; STS §B
SI-14	FW-401, FW-402, FW-403, AD-401, AD-403	UTS-CSI-010-14	ITS §10; STS §B
SI-15	FW-103, FW-604, FW-605	UTS-CSI-010-15	ITS §11; STS §B
SI-20	AD-101, AD-102, AD-103, AD-104, AD-105, AD-106, AD-107, AD-305	UTS-CSI-010-20	ITS §12; STS §C
SI-21	AD-201, AD-202, AD-203, AD-204, AD-205, AD-301, AD-302, AD-303, AD-304, AD-305, AD-401, AD-402, AD-403, AD-404, AD-405, AD-501, AD-502, AD-503, AD-504, AD-505, AD-601, AD-901, AD-902, AD-903, AD-904, AD-905	UTS-CSI-010-21	ITS §13; STS §C
SI-22	AD-601, AD-602, AD-603, AD-701, AD-702, AD-703, AD-801, AD-802	UTS-CSI-010-22	ITS §14; STS §C
SI-30	DM-101, DM-102, DM-105, DM-203, DM-301, DM-304, DM-401, DM-402, DM-403, DM-404, DM-501, DM-502, DM-503, CM-202, CM-203, CM-304	UTS-CSI-010-30	ITS §15; STS §D
SI-31	RA-307, RA-705, RA-706, RA-904, AD-902, DM-103, DM-104, DM-105, DM-504	UTS-CSI-010-31	ITS §16; STS §D
SI-32	AD-103, AD-701, AD-802, DM-201, DM-202, DM-204, DM-302, DM-303, DM-304, DM-503	UTS-CSI-010-32	ITS §17; STS §D
SI-40	RA-804, AD-801, DM-204, CM-101, CM-102, CM-103	UTS-CSI-010-40	ITS §18; STS §E
SI-41	RA-303, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, CM-201, CM-302	UTS-CSI-010-41	ITS §19; STS §E

Totals: 179 software requirements; 94 parent requirements and hazards; 19 software items. The complete requirement-level matrix, with test case identifiers and results, is exported from the requirements tool with each release and filed in the design history file.

12. Reference Documents

12.1 External standards and regulations

Reference	Title
IEC 62304:2006 + A1:2015	Medical device software: software life cycle processes
IEC 82304-1	Health software: general requirements for product safety
ISO 14971:2019	Medical devices: application of risk management to medical devices
IEC 62366-1	Medical devices: application of usability engineering to medical devices
21 CFR Part 11	Electronic records; electronic signatures
21 CFR Part 820 (QMSR)	Quality Management System Regulation
FDA premarket cybersecurity guidance	Cybersecurity in medical devices: quality system considerations and content of premarket submissions
IEC 81001-5-1	Health software and health IT systems: security activities in the product life cycle
ISAC FCS 3.1	Flow Cytometry Standard data file format
HL7 v2	Health Level Seven messaging standard (LIS export)

12.2 Program documents

Document no.	Rev	Title
SRS-CSI-001	D	System Requirements Specification
SAD-CSI-001	C	System Architecture Description
SAD-SW-CSI-011	C	Software Architecture Description
SDS-SW-CSI-012	B	Application Detailed Design Specification and User Interface Style Guide
SDS-FW-CSI-013	B	Firmware and FPGA Detailed Design Specification (including board register map)
SDP-CSI-010	C	Software Development Plan
SCMP-CSI-010	B	Software Configuration Management Plan
SVP-CSI-010	B	Software Verification Plan
UTS-CSI-010-01 to -41	A	Unit Test Specifications (one per software item)
ITS-CSI-010	A	Software Integration Test Specification
STS-CSI-010	A	Software System Test Specification
SVR-SW-CSI-010	A	Software Verification Report
ICD-CSI-010	B	Interface Control Document: Sorter to Fluidics Supply Module
ICD-CSI-020	A	Interface Control Document: Sorter to Workstation
CFG-CSI-001	C	Configuration Definition: Laser and Detector Builds
SSD-CSI-600	B	Electronics and Signal Processing Subsystem Design Specification
SSD-CSI-700	C	Sort Subsystem Design Specification
FTS-CSI-001	A	Factory Alignment and Test Specification
HA-CSI-001	C	Hazard Analysis
RMP-CSI-001	B	Risk Management Plan
OG-CSI-001	B	Operator's Guide (Instructions for Use)
SM-CSI-001	A	Service Manual
DTP-CSI-001	B	Design Transfer Plan

Appendix A. Terms and Abbreviations

Term	Meaning
As-found / as-left	Measured value before / after a calibration adjustment
Drop 1 / Gap	Breakoff image metrics: position of the first free drop, and gap between the stream end and the first drop
Drop delay	Time, in drops, from interrogation to breakoff; determines which drop is charged
FPGA / CPLD	Field-programmable gate array / complex programmable logic device
HA	Hazard identified in HA-CSI-001
HV	High voltage (deflection plates or PMT supply)
LIS	Laboratory information system
Mask (yield, purity, phase)	Rules, in 1/32-drop steps, deciding whether a drop is sorted when events are near drop edges
RTOS	Real-time operating system
Safe state	Instrument state defined in Section 5.6
SBOM	Software bill of materials
SI-nn	Software item identifier (Section 3.2)
Software safety class	Class A, B or C per IEC 62304 4.3
SOUP	Software of unknown provenance (IEC 62304 3.29)
RA / FW / AD / DM / CM	Requirement prefixes: Research Application, Firmware, Admin Application, Data Management Application, Common